Temperature Sensor Project

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Introduction

For this document my lab partner and I will discuss how to create a temperature sensor. The reader will learn the design required for a temperature sensor using electronic systems built by circuits, what the system should look like physically on a breadboard, what struggles can occur during the process, as well as what we had to face during our designing process. For this project, the goal is to make a Temperature sensor that has different color lights for different variations of heat that the system will recognize and give a response to via light emitting diode. The project has some restrictions; The input voltage is controlled by a heat sensor, which gives a certain amount of voltage, depending on the temperature, in Celsius. Other restrictions include that the green light has to come on first, when the system is below 40 degrees (Celsius). When the temperature is above 40 degrees, the green light turns off and the orange light turns on. Then, when the temperature gets above 100 degrees the orange light will remain on, however, the red LED will turn on as well. When decreasing the temperature of the system, the temperature has to get below 80 degrees for the red light to turn off. Then when it is below 30 degrees the orange light will turn off and the green light will turn back on. Overall, the project has a few concepts that need understanding before beginning the process of how to build this system. Some of the concepts that will be needed are op amp configuration and comparator op amps. In the Bibliography, there will be some great resources from Texas Instruments, and Professors of Iowa State University that my lab partner and I used to build this circuit.

List of Items

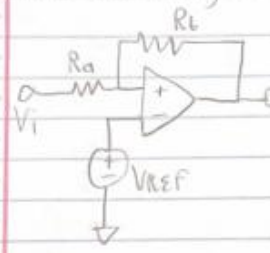
- Breadboard
- 230 lab kit materials (wire, resistor, potentially other passive components)
- The LMC 660 Op-amp
- LT-Spice
- Lab equipment (function generator, multimeter, etc.)

Circuit Calculations

EE-230 project: Temp sensor

In order to get LEDs to turn off/on at certain voltages, use comparator.

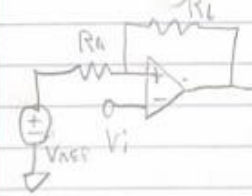
Non-inverting comparator



If $V_t > V_{REF}$, $V_o = V_{s-}$
If $V_t < V_{REF}$, $V_o = V_{s+}$

$$V_{TL} = V_{REF}\left(1 + \frac{R_a}{R_f}\right) - \left(\frac{R_a}{R_f}\right) V_{L+}$$
$$V_{TH} = V_{REF}\left(1 + \frac{R_a}{R_f}\right) - \left(\frac{R_a}{R_f}\right) V_{L-}$$

Inverting comparator



If $V_t > V_{REF}$, $V_o = V_{s+}$
If $V_t < V_{REF}$, $V_o = V_{s-}$

$$V_{TH} = \frac{V_{REF} + \left(\frac{R_a}{R_f}\right) V_{L+}}{1 + \frac{R_a}{R_f}}$$

$$V_{TL} = \frac{V_{REF} + \left(\frac{R_a}{R_f}\right) V_{L-}}{1 + \frac{R_a}{R_f}}$$

Requirements

Temp increasing:
- At 40°C (0.4V), green turns off, yellow turns on,
- At 100°C (1V), red turns on, yellow stays on

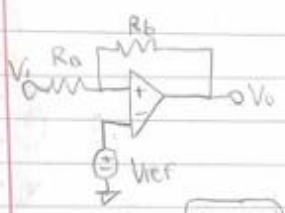
Temp decreasing:
- At 80°C (0.8V), red turns off
- At 30°C (0.3V), yellow turns off, green turns back on

That means, green will be an inverting comparator as, it needs to stay on until a certain voltage, and the orange and yellow are non-inverting as they won't turn on until a certain voltage

Adding a op-amp to raise voltage, A=10, multiply every V_{TL}/V_{TH} by 10.

Green - $V_{TH} = 0.4 \cdot 10 = 4V$, $V_{TL} = 0.3 \cdot 10 = 3V$ $V_{cc+} = V_{L+} = 12V$
Orange - $V_{TH} = 0.4 \cdot 10 = 4V$, $V_{TL} = 0.3 \cdot 10 = 3V$ $V_{cc-} = V_{L-} = -12V$
Red - $V_{TH} = 1 \cdot 10 = 10V$, $V_{TL} = 0.8 \cdot 10 = 8V$

Yellow calculations



$$V_{TH} - V_{TL} = \frac{R_a}{R_b}(V_{L+} - V_{L-})$$

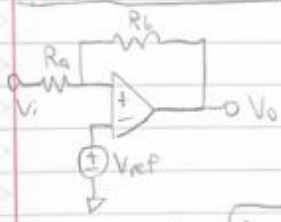
$$\frac{R_a}{R_b} = \frac{V_{TH} - V_{TL}}{V_{L+} - V_{L-}} = \frac{4-3}{12+12} = \frac{1}{24}$$

$$V_{REF} = \frac{V_{TL} + \frac{R_a}{R_b} V_{L+}}{1 + \frac{R_a}{R_b}} = \frac{3 + (\frac{1}{24})12}{1 + (\frac{1}{24})} = \boxed{3.48V}$$

Say $\boxed{R_a = 1k\Omega}$ → $4 = 3.48(1 + \frac{1k}{R_b}) - (\frac{1k}{R_b})(-12) = 3.48 + \frac{15.48k}{R_b}$

$0.52R_b = 15480$ → $\boxed{R_b = 29.8K\Omega \approx 30K\Omega}$

Red calculations



$$V_{TH} - V_{TL} = \frac{R_a}{R_b}(V_{L+} - V_{L-})$$

$$\frac{R_a}{R_b} = \frac{V_{TH} - V_{TL}}{V_{L+} - V_{L-}} = \frac{10-8}{12+12} = \frac{1}{12}$$

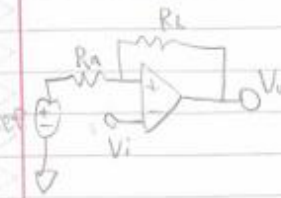
$$V_{REF} = \frac{V_{TL} + \frac{R_a}{R_b} V_{L+}}{1 + \frac{R_a}{R_b}} = \frac{8 + (\frac{1}{12})12}{1 + \frac{1}{12}} = \boxed{8.31V}$$

Say $\boxed{R_a = 1k\Omega}$ → $10 = 8.31(1 + \frac{1k}{R_b}) - (\frac{1k}{R_b})(-12)$

$10 = 8.31 + \frac{8.31k}{R_b} + \frac{12k}{R_b} = 8.31 + \frac{20.31k}{R_b}$

$1.69R_b = 20310$ → $\boxed{R_b = 12K\Omega}$

Green calculations



$$V_{TH} - V_{TL} = \frac{R_a}{R_b}(V_{L+} - V_{L-})$$

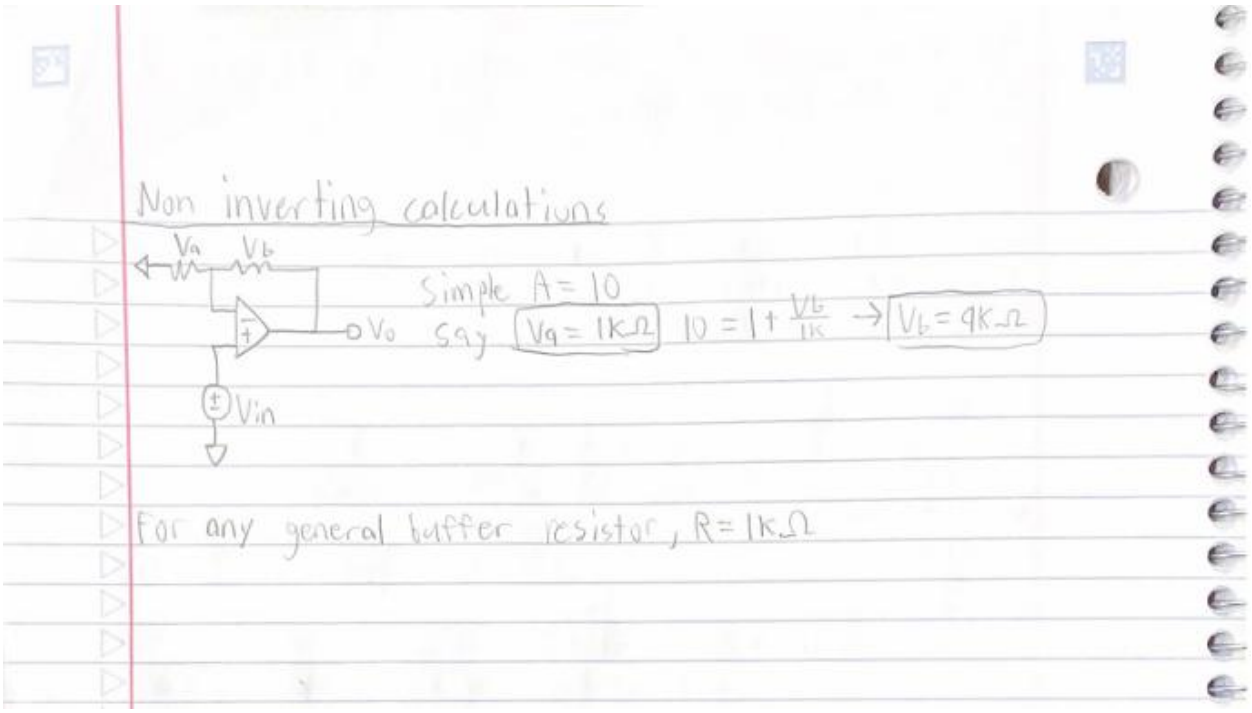
$$\frac{R_a}{R_b} = \frac{V_{TH} - V_{TL}}{V_{L+} - V_{L-}} = \frac{4-3}{12+12} = \frac{1}{24}$$

$$V_{REF} = V_{TH}(1 + \frac{R_a}{R_b}) - (\frac{R_a}{R_b})V_{L+} = 4(1 + \frac{1}{24}) - (\frac{1}{24})12$$

$$\boxed{V_{REF} = 4.167 - 0.5 = 3.67V}$$

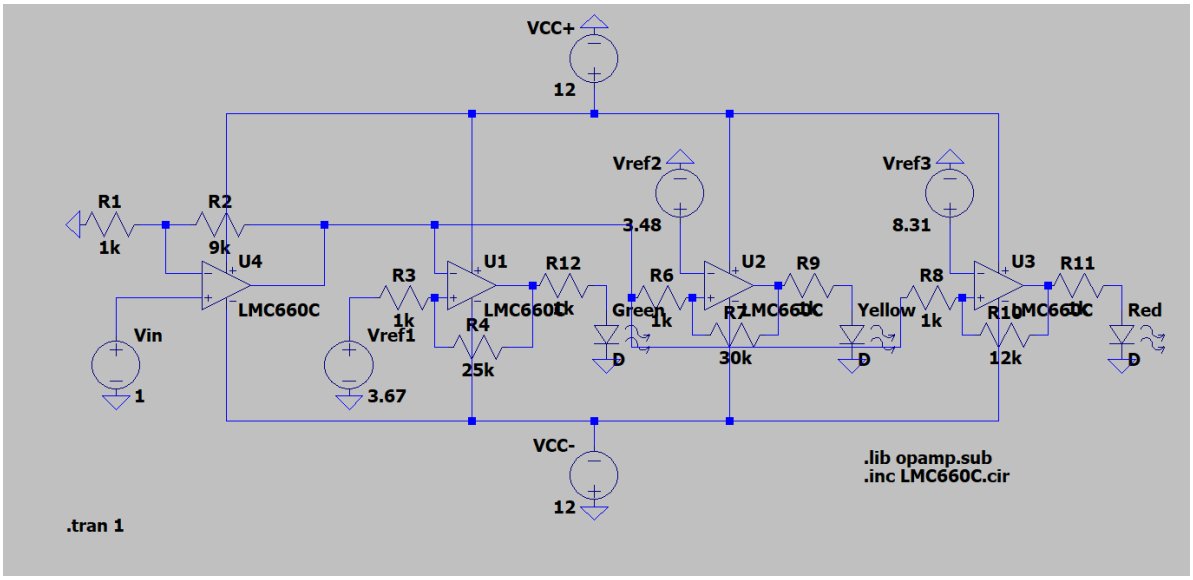
Say $\boxed{R_a = 1k\Omega}$ → $4 = \frac{3.67 + \frac{1k}{R_b}(12)}{1 + \frac{1k}{R_b}}$ → $4 + \frac{4k}{R_b} = 3.67 + \frac{12k}{R_b}$

$0.33 = \frac{8k}{R_b}$ → $\boxed{R_b = 24.2k\Omega \approx 25k\Omega}$



Circuit Schematic

The buffer resistors on the diode nodes are all 1k ohm.

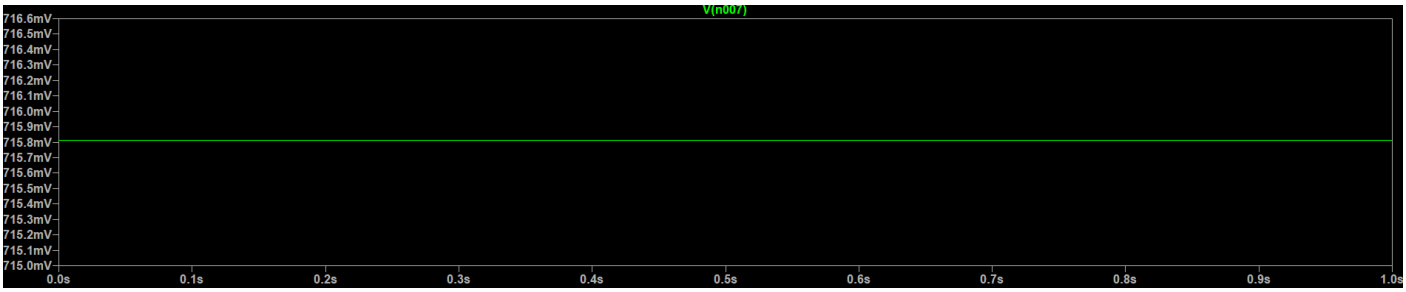


Simulation Results

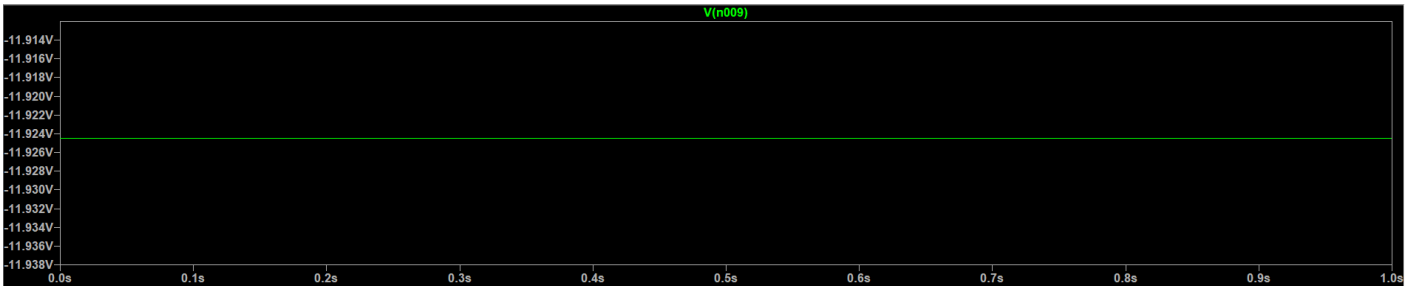
For reference, the diode is on when at 0.7V and off when at –12V.

0.2V (20 degrees Celsius)

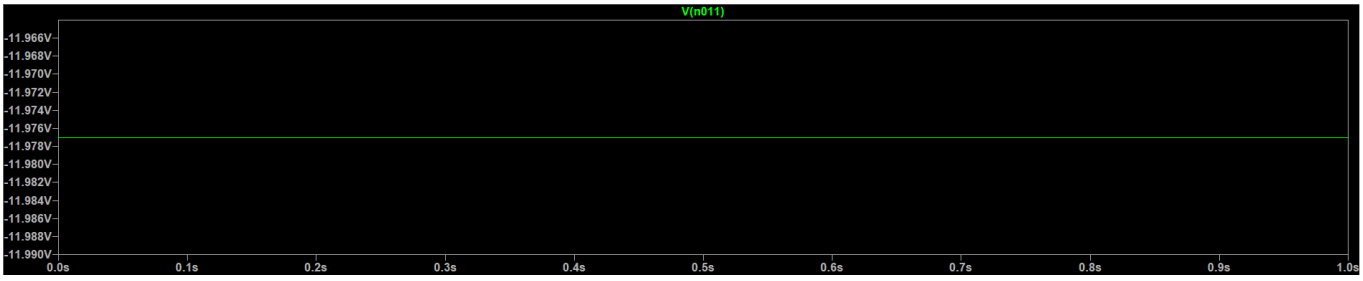
Green LED



Orange LED

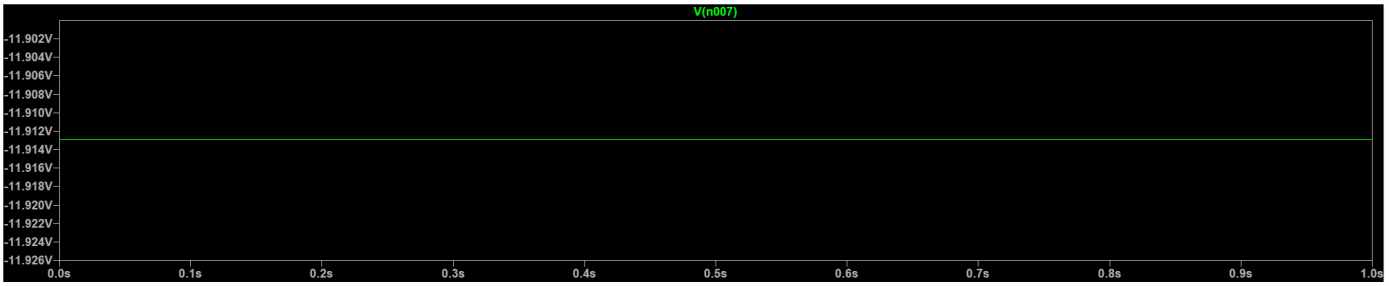


Red LED

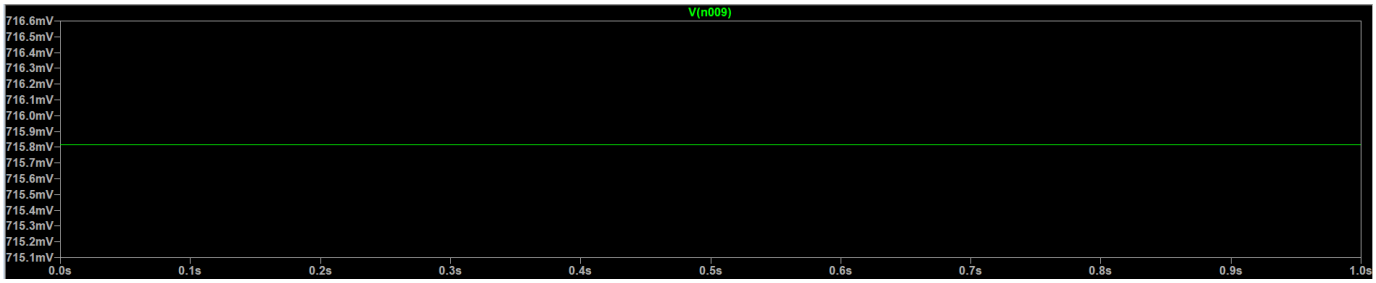


0.4V (40 degrees Celsius)

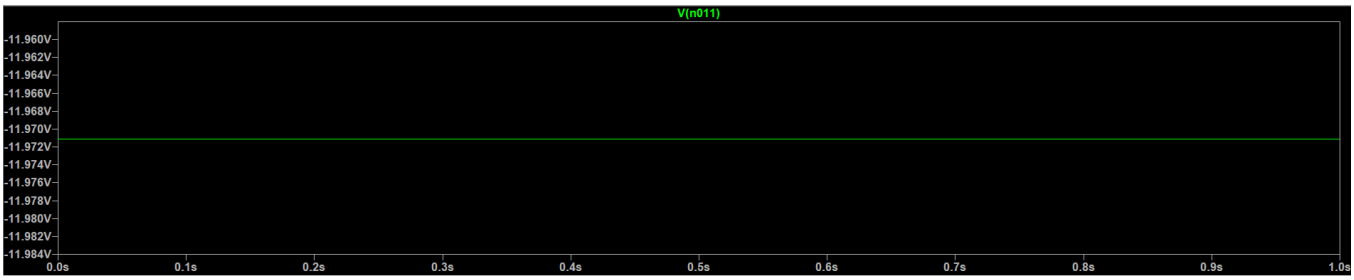
Green LED



Orange LED

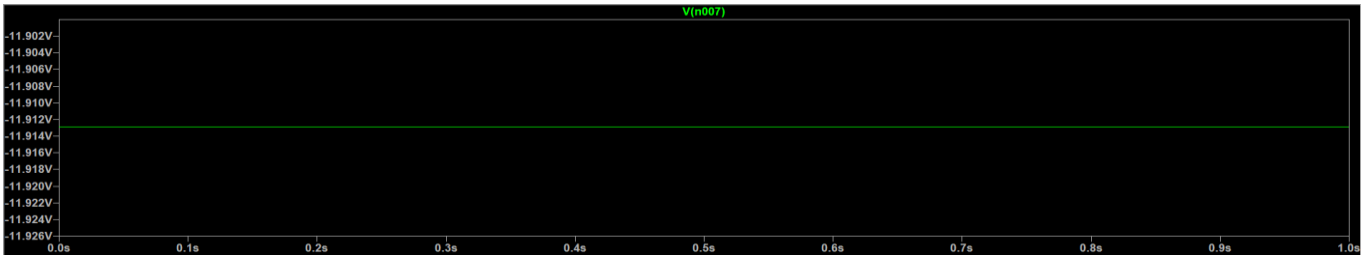


Red LED

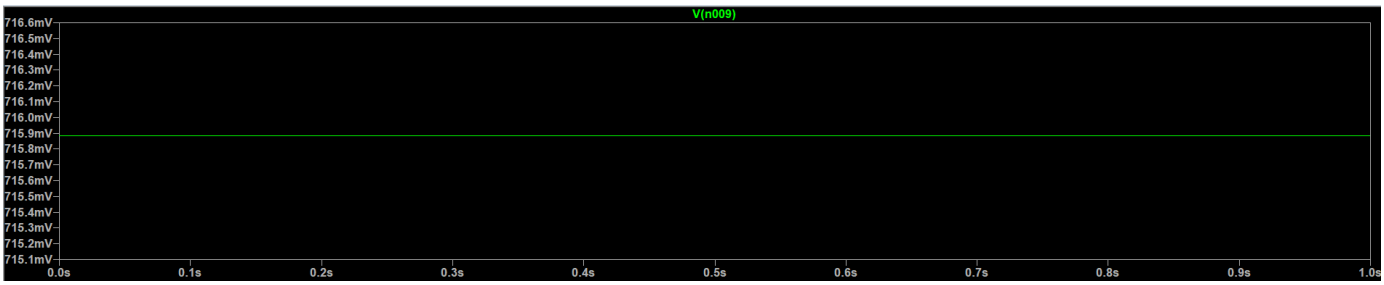


1V (100 degrees Celsius)

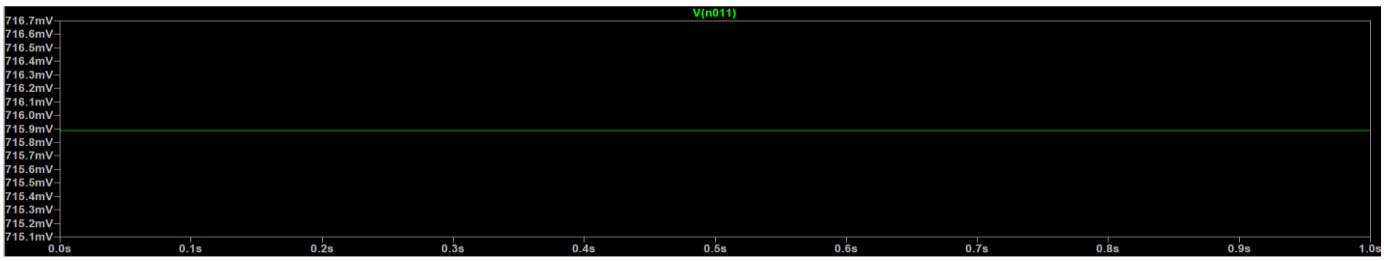
Green LED



Orange LED



Red LED



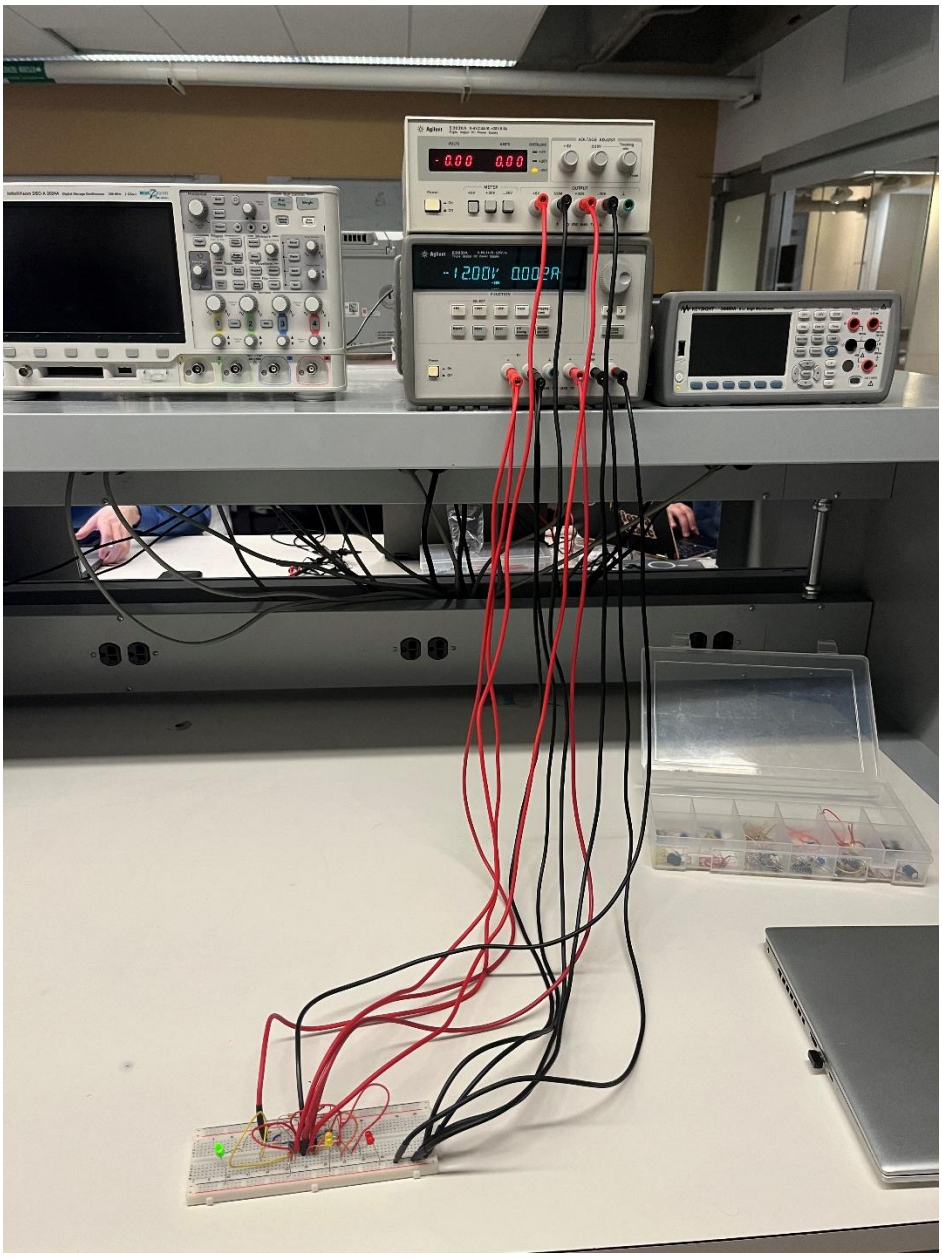
Lab Results

To do the Temperature sensor the user would have to build a similar circuit to ours. In this case, as the reader can see, my lab partner and I did not use the temp sensor component when making the circuit. We started out with a normal non-inverting op-amp. This was used to amplify the voltage source to get the rest of the circuit to begin the temp sensing of the system. The next 3 op-amps are also variations of Hysteresis comparators. When designing our circuit, we ran into many issues along the way. At first, we tried using Hysteresis comparators with transistors as switches to make the circuit. We found out after many tries that we were not building the circuits correctly with the transistors, so the TA's told us to rebuild. The reason why we are using Hysteresis comparators is because they are used to decipher when to do a task based off the range of the system. So, in this case, using this style of amplifier will let the circuit know when to turn the light off or on. Unfortunately, we did not meet some of the criteria for our results of the lab. In the lab document that is provided, it discusses how the designer needs to make the circuit work in such a way that when the temperature changes from the threshold conditions, the lights will either turn on or off. For our case, this is where my lab partner and I's problems came in. When we tried to reference the voltage to each op-amp. We did not use voltage dividers like we were supposed to. This along with some off resistor values caused some error in our final product. The circuit itself worked, however, when the lights were supposed to turn on and off, it was a little bit off in terms of temperature.

Conclusion

Overall, this lab was enjoyable. It was fun working with diodes and being able to create something of our own rather than just going off a lab document. Our issues came in when trying to get the sensor to be a 100 percent perfect sensor, however, our achievements were through not giving up, making a decent design, and getting the sensor to work. In all, the lab was fun and we hope to do something like this again in the future.

Circuit Images

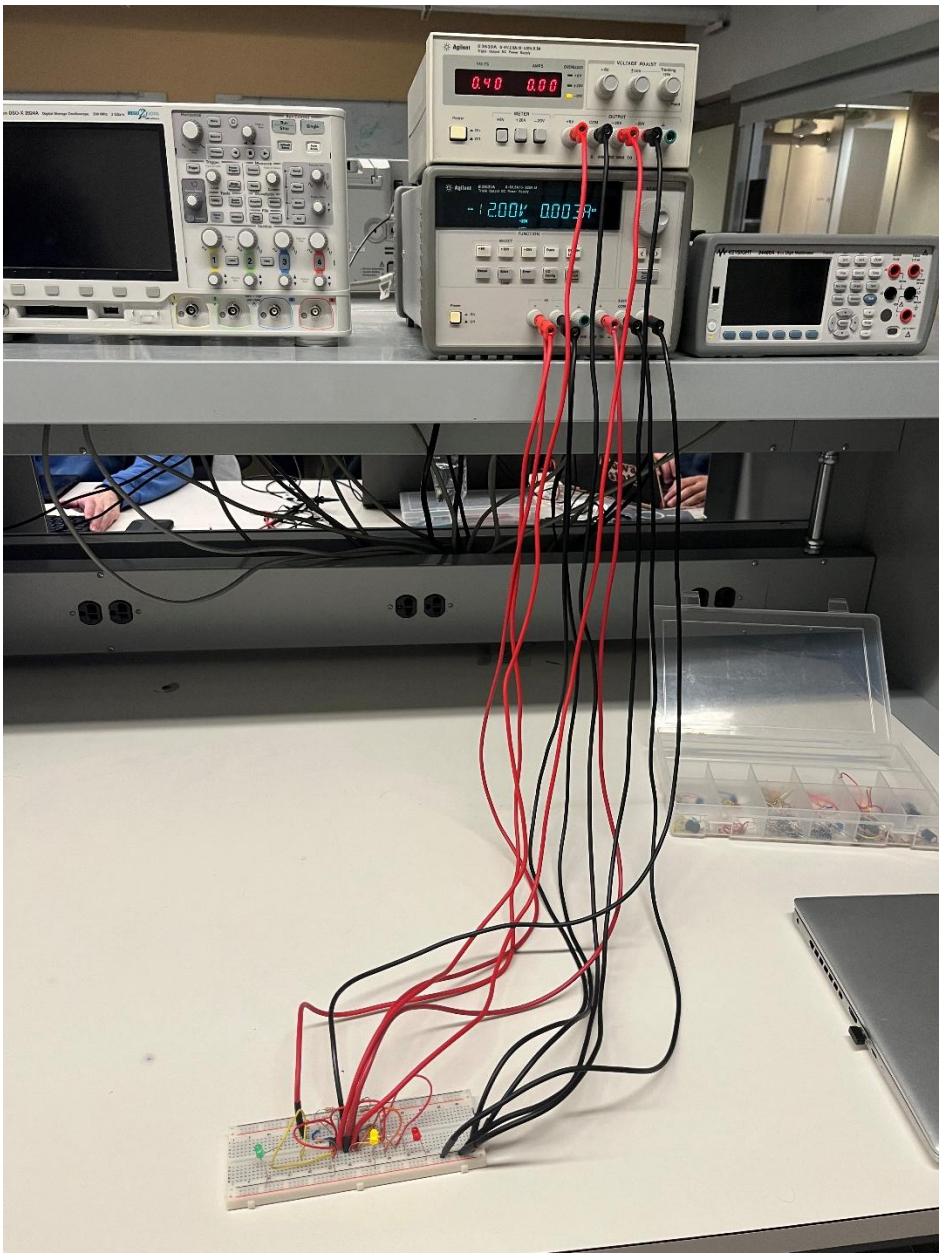


Green light: on

Orange light: off

Red light: off

Voltage: 0 V

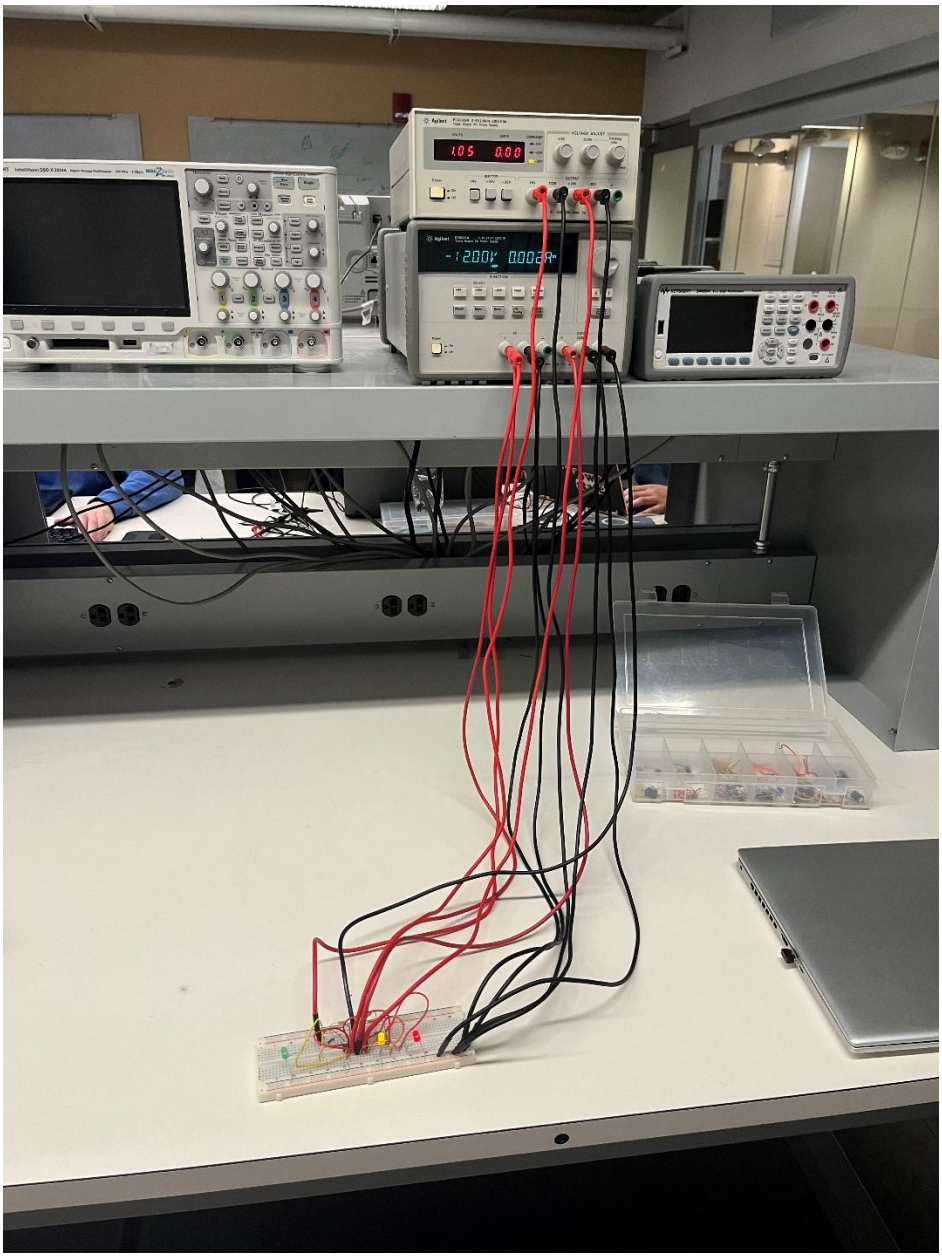


Green light: off

Orange light: on

Red light: off

Voltage: 0.4 V

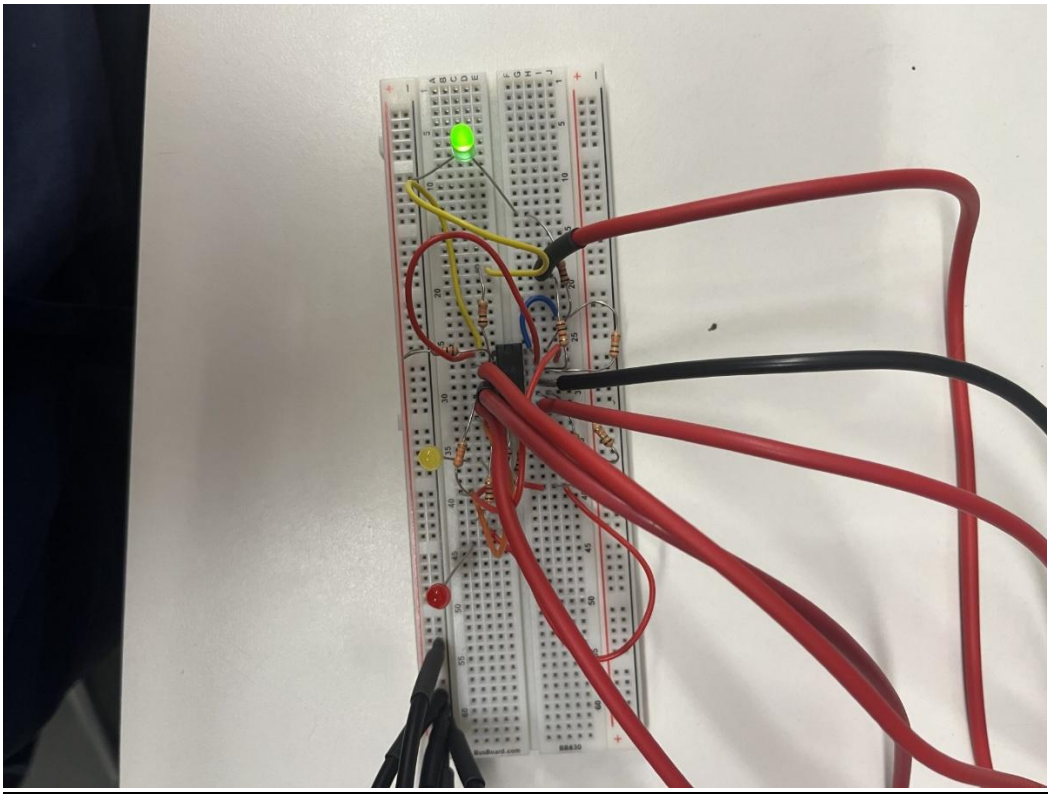


Green light: off

Orange light: on

Red light: on

Voltage: 1.05 V



Bibliography

Tuttle, Gary. *Comparators*, 16 Apr. 2025, gtuttle.net/electronics/topics/comparators.pdf

Kay, Art, and Timothy Claycomb. *TI Designs -Precision: Verified Design Comparator with Hysteresis Reference Design,* 2013, Comparator with Hysteresis Reference Design